

Module 3: IAM Policies Assignment

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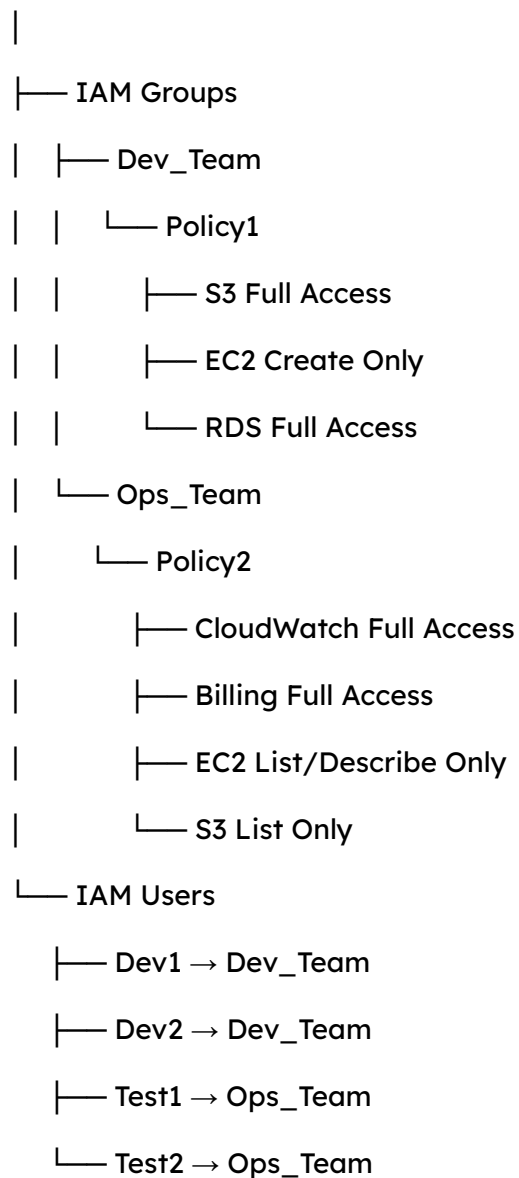
Problem Statement

You work for XYZ Corporation. To maintain the security of the AWS account and the resources, a solution is required that helps easily recognize and monitor the different users.

The assignment implements this through IAM customer-managed policies and group-based access control.

FlowChart

AWS Account

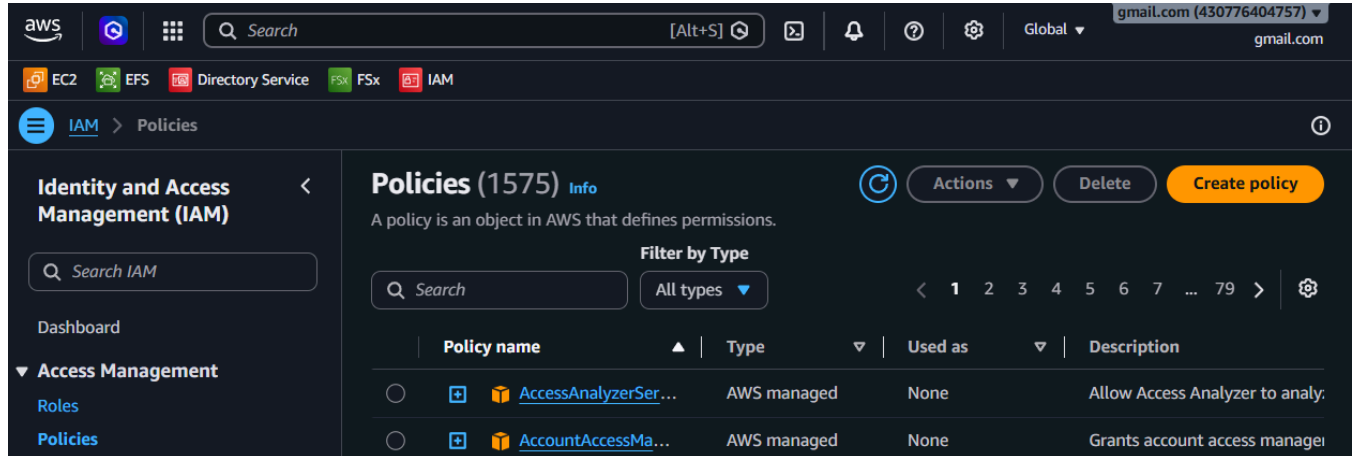


Task 1 – To create Policy_1

To create Policy_1 with complete S3 access, permission only to create EC2 instances, and full RDS access.

Steps

1. AWS Console → IAM → Policies → Create policy



2. Specified Policies.

Service	Permission
S3	Full access
EC2	Create/launch instances only
RDS	Full access

The screenshot shows the AWS IAM console interface for creating a new policy. The breadcrumb navigation is IAM > Policies > Create policy. The left sidebar shows a two-step process: Step 1 'Specify permissions' (selected) and Step 2 'Review and create'. The main area is titled 'Specify permissions' with an 'Info' link. Below the title is a description: 'Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.' The 'Policy editor' section has tabs for 'Visual' (selected), 'JSON', and 'Actions'. It lists three selected services: S3 (Allow, All actions), EC2 (Allow, 1 Action), and RDS (Allow, All actions). Each service entry has a plus icon to add more and a trash icon to remove it. At the bottom, there is a '+ Add more permissions' button and a status bar showing 'Security: 0', 'Errors: 0', 'Warnings: 0', and 'Suggestions: 2'. The bottom right corner has 'Cancel' and 'Next' buttons.

aws [Search] [Alt+S] Global gmail.com (430776404757) gmail.com

EC2 EFS Directory Service FSx IAM

IAM > Policies > Create policy

Step 1
Specify permissions
Step 2
Review and create

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor Visual JSON Actions

- ▶ **S3**
Allow All actions
- ▶ **EC2**
Allow 1 Action
- ▶ **RDS**
Allow All actions

+ Add more permissions

Security: 0 Errors: 0 Warnings: 0 Suggestions: 2

Cancel Next

3. The policy name was entered as **Policy_1** and a description was added.

The screenshot shows the AWS IAM console interface for creating a new policy. The breadcrumb navigation at the top indicates the path: IAM > Policies > Create policy. On the left, a progress bar shows two steps: 'Specify permissions' (Step 1) and 'Review and create' (Step 2, which is the active step). The main content area is titled 'Review and create' with an 'Info' link. Below the title, it says 'Review the permissions, specify details, and tags.' The 'Policy details' section contains two text input fields. The first is for the 'Policy name', which has 'Policy_1' entered. Below it, a note states: 'Maximum 128 characters. Use alphanumeric and '+=, @-_' characters.' The second field is for the 'Description - optional', which has 'Access S3 completely, Only create EC2 instances, Full access to RDS' entered. Below this, a note states: 'Maximum 1,000 characters. Use alphanumeric and '+=, @-_' characters.' The 'Permissions defined in this policy' section has an 'Edit' button. It includes a search bar and a toggle for 'Show remaining 472 services'. Below this is a table with three columns: 'Service', 'Access level', and 'Resource'. The table lists three services: EC2, RDS, and S3. EC2 has 'Limited: Write' access to 'All resources'. RDS has 'Full access' to 'All resources'. S3 has 'Full access' to 'All resources'.

Step 1
Specify permissions

Step 2
Review and create

Review and create [Info](#)

Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

Policy_1

Maximum 128 characters. Use alphanumeric and '+=, @-_' characters.

Description - optional
Add a short explanation for this policy.

Access S3 completely, Only create EC2 instances, Full access to RDS

Maximum 1,000 characters. Use alphanumeric and '+=, @-_' characters.

Permissions defined in this policy [Info](#) [Edit](#)

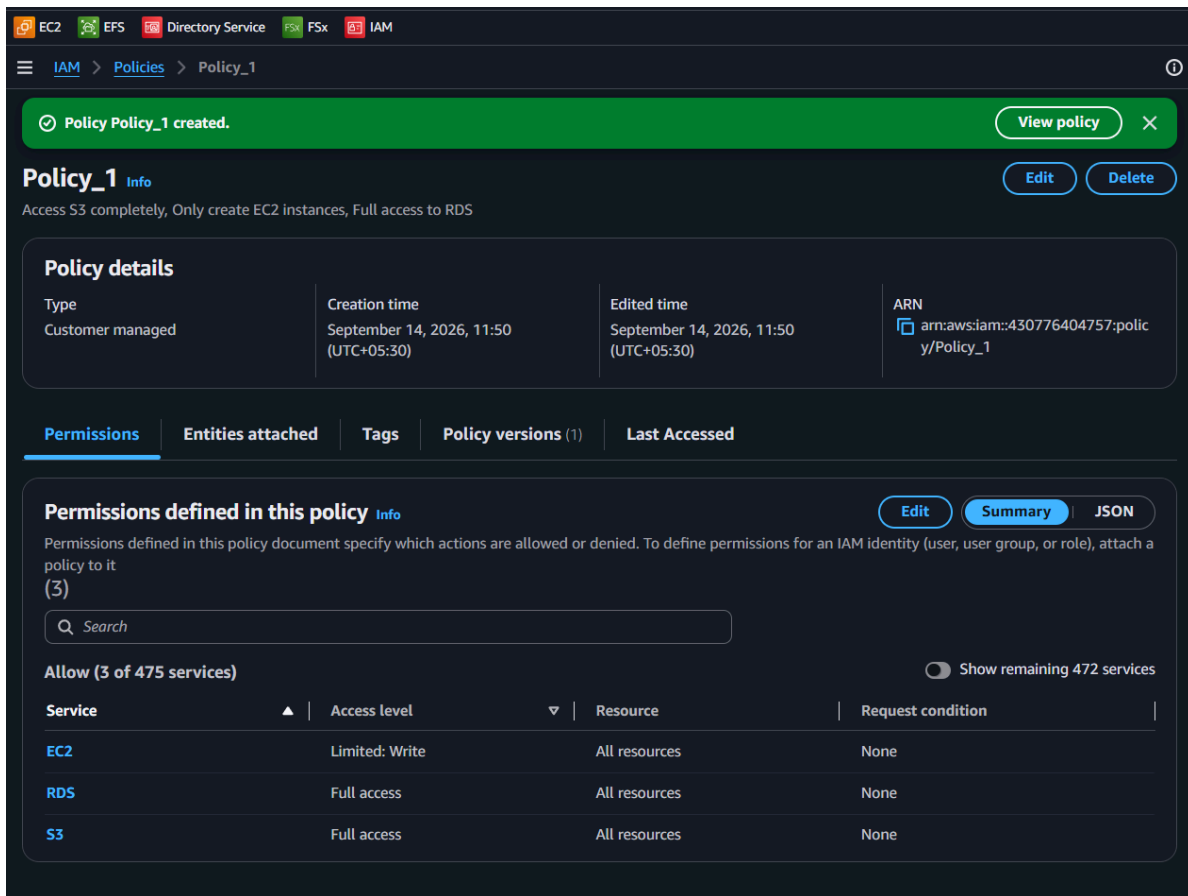
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

(3)

Allow (3 of 475 services) ☐ Show remaining 472 services

Service	Access level	Resource
EC2	Limited: Write	All resources
RDS	Full access	All resources
S3	Full access	All resources

4. The Policy_1 was created and the successful creation message displayed by AWS was verified.



Task 2 – To Create Policy_2

To create Policy_2 with complete CloudWatch and billing access, while allowing only EC2 and S3 resource listing.

Steps

1. IAM → Policies → Create policy → JSON
2. Selected the **JSON** tab and used:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "CloudWatchFullAccess",
      "Effect": "Allow",
      "Action": "cloudwatch:*",
      "Resource": "*"
    }
  ],
  {
```

```

        "Sid": "BillingFullAccess",
        "Effect": "Allow",
        "Action": [
            "aws-portal:*",
            "billing:*",
            "budgets:*",
            "ce:*"
        ],
        "Resource": "*"
    },
    {
        "Sid": "EC2ListOnly",
        "Effect": "Allow",
        "Action": [
            "ec2:Describe*"
        ],
        "Resource": "*"
    },
    {
        "Sid": "S3ListOnly",
        "Effect": "Allow",
        "Action": [
            "s3:ListAllMyBuckets",
            "s3:GetBucketLocation"
        ],
        "Resource": "*"
    }
]
}

```

Service	Permission
S3	Full access
EC2	Create/launch instances only
RDS	Full access

aws

Search

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Global

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gmail.com

EC2

EFS

Directory Service

FSx

IAM

IAM

Policies

Create policy

Step 1

Specify permissions

Step 2

Review and create

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

VisualJSONActions

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34

```
"Version": "2012-10-17",
"Statement": [
  {
    "Sid": "FullCloudWatchAccess",
    "Effect": "Allow",
    "Action": "cloudwatch:*",
    "Resource": "*"
  },
  {
    "Sid": "FullBillingAccess",
    "Effect": "Allow",
    "Action": [
      "aws-portal:*",
      "billing:*",
      "budgets:*",
      "ce:*",
      "cur:*",
      "purchase-orders:*"
    ],
    "Resource": "*"
  },
  {
    "Sid": "ListOnlyEC2Access",
    "Effect": "Allow",
    "Action": [
      "ec2:Describe*",
      "ec2:List*",
      "ec2:Get*"
    ],
    "Resource": "*"
  },
  {
    "Sid": "ListOnlyS3Access",
```

49:2JSON

+ Add new statement

5508 of 6144 characters remaining

Security: 0Errors: 0Warnings: 0Suggestions: 0

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

Cancel

Next

3. The policy name was entered as **Policy_2** and a description was added.

The screenshot shows the 'Review and create' step in the AWS IAM console. The policy name is 'Policy_2' and the description is 'Access CloudWatch and billing completely, Can only list EC2 and S3 resources'. The permissions section shows 9 services selected: Billing, Billing Console, Budget, CloudWatch, Cost and Usage Report, Cost Explorer Service, EC2, Purchase Orders, and S3.

Policy details

Policy name
Enter a meaningful name to identify this policy.
Policy_2
Maximum 128 characters. Use alphanumeric and '+=, @-_' characters.

Description - optional
Add a short explanation for this policy.
Access CloudWatch and billing completely, Can only list EC2 and S3 resources
Maximum 1,000 characters. Use alphanumeric and '+=, @-_' characters.

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it (9)

Search

Allow (9 of 475 services) Show remaining 466 services

Service	Access level	Resource	Request condition
Billing	Full access	All resources	None
Billing Console	Full access	All resources	None
Budget	Full access	All resources	None
CloudWatch	Full access	All resources	None
Cost and Usage Report	Full access	All resources	None
Cost Explorer Service	Full access	All resources	None
EC2	Limited: List, Read	All resources	None
Purchase Orders	Full access	All resources	None
S3	Limited: List, Read	All resources	None

4. Policy_2 was created.

The screenshot shows the 'Policies' page in the AWS IAM console. It lists two policies: Policy_1 and Policy_2. Policy_2 is highlighted.

Policies (1577)

A policy is an object in AWS that defines permissions.

Filter by Type: Customer managed (2 matches)

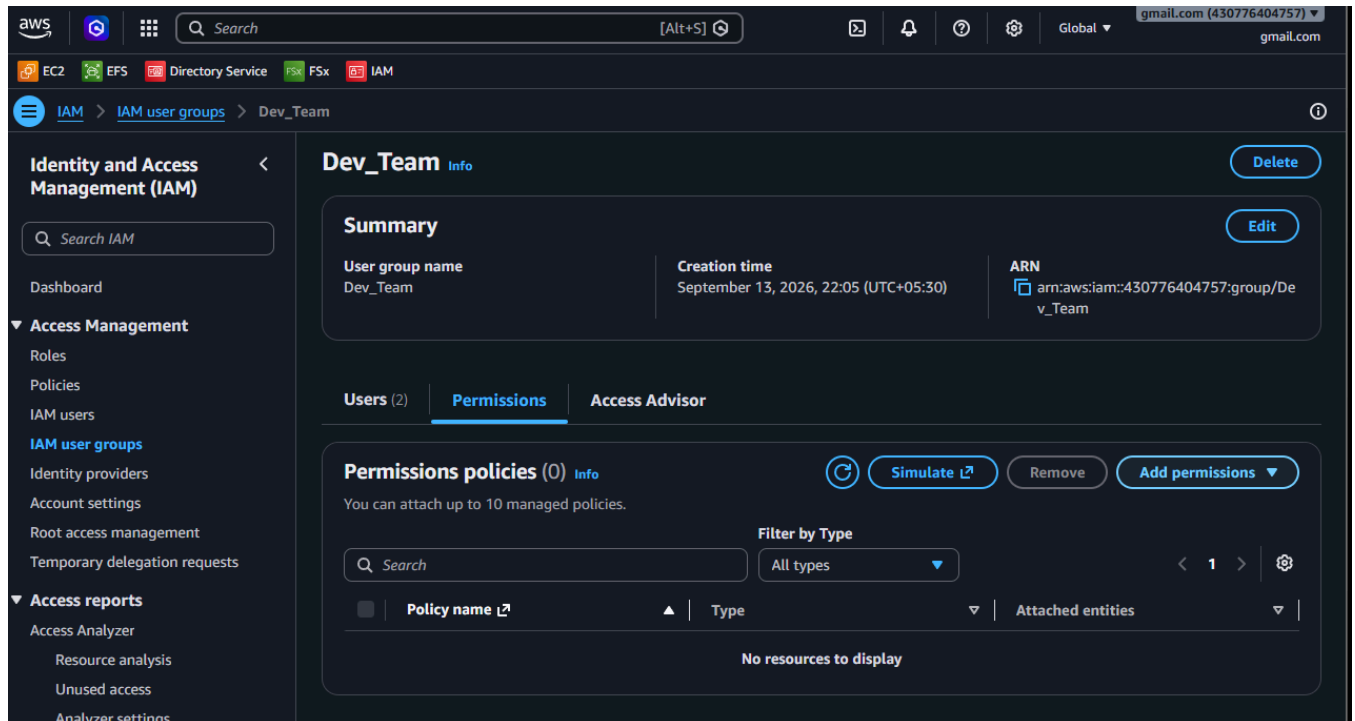
Policy name	Type	Used as	Description
Policy_1	Customer managed	None	Access S3 completely, Only create EC2 ...
Policy_2	Customer managed	None	Access CloudWatch and billing comple...

Task 3 – Attach Policy_1 to Dev_Team

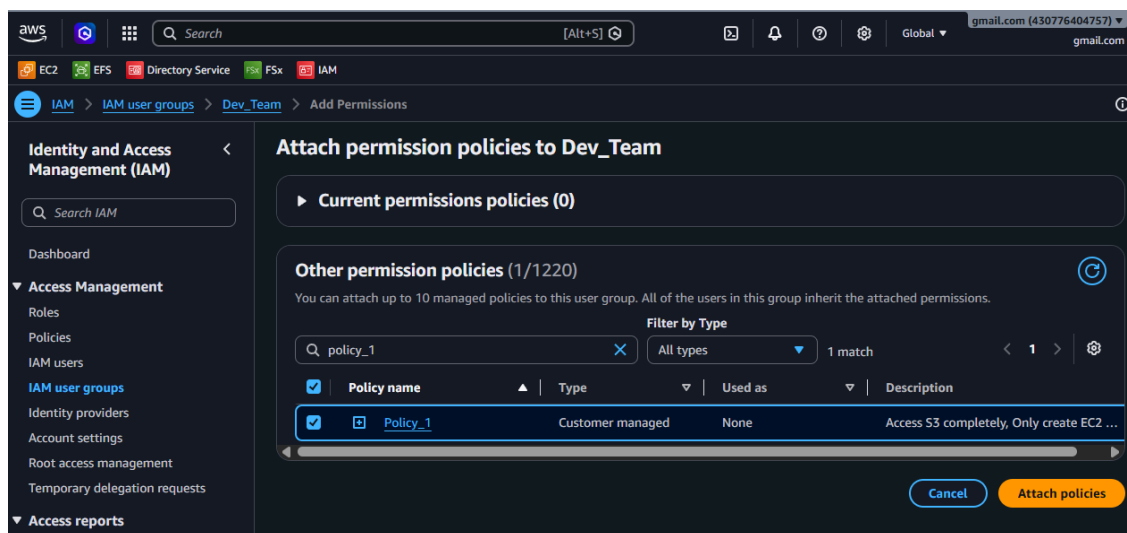
To Attach Policy_1 to the Dev_Team group.

Steps

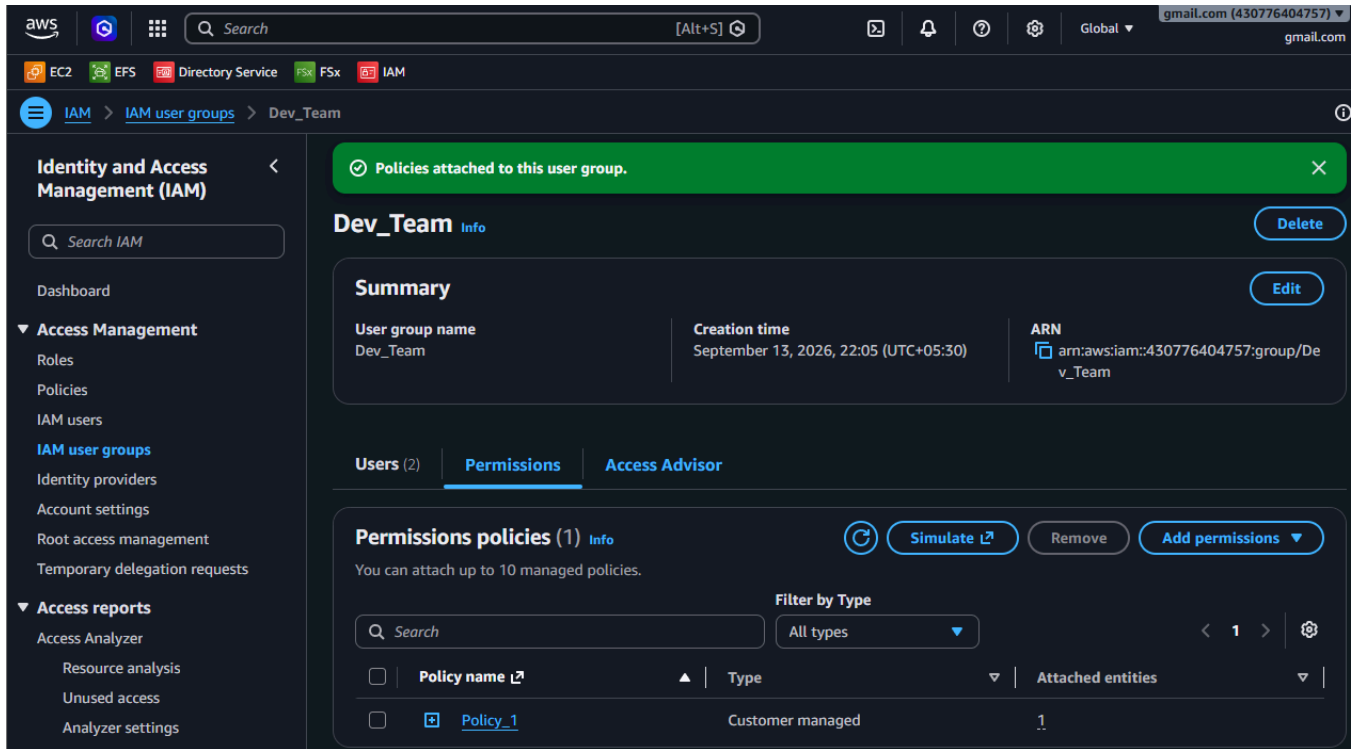
1. IAM → User groups → Dev_Team .



2. Permissions → Add permissions → Attach policies.
3. **Policy_1** selected and clicked **Add permissions**.



4. Policy_1 attached to Dev_Team



Task 4: Attach Policy1 to Dev_Team

To Attach Policy_2 to the Ops_Team group.

Steps

1. IAM → User groups → Ops_Team was opened.
2. Permissions → Add permissions → Attach policies.
3. Policy_2 was selected and attached to Ops_Team.

